Radionuclide Identification for Plastic Scintillation Detectors by Backward Cumulative Channel Sum

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Abstract

Large-area plastic scintillation detectors are widely used in radiation portal monitors, but their Compton-dominated spectra provide limited radionuclide-specific information. We evaluated backward cumulative channel sum (BCCS) combined with non-negative least-squares (NNLS) decomposition and compared it with raw-spectrum NNLS using an explicit background template and detector-specific five-window NNLS. A measured background and fixed reference spectra for Am-241, Cs-137, and Co-60 were used to analyze 18 independent spectra: nine single-radionuclide measurements, eight binary mixtures, and one ternary mixture. The mixed-source compositions were concealed during the primary BCCS analysis. All three methods ranked the actual radionuclide first for every single-radionuclide spectrum and the two constituents first and second for every binary mixture; the ternary mixture yielded positive coefficients for all three radionuclides. For the nine mixed-source spectra, method-specific relative residuals were 0.48–1.31% for BCCS-NNLS, 0.88–1.60% for raw-spectrum NNLS, and 0.46–2.57% for five-window NNLS. Parametric Poisson resampling supported stable coefficient rankings under counting-statistical variation. These results support BCCS-NNLS as a simple, linear, and interpretable approach for static, unshielded measurements within a fixed radionuclide library, without hardware modification or data-intensive training. The study does not address open-set identification, activity quantification, or formal detection limits.

Keywords: plastic scintillator; radiation portal monitor; radionuclide identification; backward cumulative channel sum; non-negative least squares; background subtraction

Introduction

Radiation portal monitors (RPMs) are widely deployed at borders, ports, transportation facilities, and industrial sites to detect unauthorized radioactive materials. Large-area plastic scintillation detectors are commonly used because of their high sensitivity, rapid response, mechanical robustness, and moderate cost [1,2]. Their principal limitation is poor energy resolution. Because gamma-ray interactions in plastic scintillators are dominated by Compton scattering, measured spectra generally consist of broad continua rather than distinct full-energy peaks, making radionuclide identification dependent on distributed spectral-shape information.

Various methods have been investigated to extract radionuclide-dependent information from such low-resolution spectra. Energy-window approaches use integrated counts or ratios within selected channel regions [3,4], whereas raw-spectrum fitting retains the full channel information. Principal-component analysis, inverse-calibration methods, energy-weighted transformations, and machine-learning or deep-learning classifiers have also been studied [5–11]. Although these approaches can improve identification performance, they may require detector-specific window optimization, calibration data, model training, or substantial computational resources. Previous studies by the authors similarly explored data-driven reconstruction, classification, and interpretation methods for plastic-scintillator spectra [11–14].

For practical RPM operation, a useful method should be computationally simple, interpretable, and suitable for direct implementation in the data-acquisition system. Under stable detector conditions, a measured spectrum may be approximated as the sum of environmental background and radionuclide-dependent detector responses. Mixed-source spectra can therefore be represented as non-negative combinations of fixed single-radionuclide references, provided that the measurement and reference conditions remain sufficiently comparable.

On this basis, we developed the Backward Cumulative Channel Sum (BCCS). At each threshold channel, BCCS integrates the counts from that channel to the highest acquired channel, converting broad spectral differences into smooth cumulative-response profiles. Because the transformation preserves linear additivity, background-subtracted mixed-source measurements can be reconstructed using non-negative least squares (NNLS) and fixed single-radionuclide reference profiles. The fitted coefficients provide detector-response scaling factors for constituent ranking rather than direct estimates of radionuclide activity. The method requires neither model training nor specialized computing hardware and is therefore potentially suitable for embedded RPM processing.

The present study evaluates BCCS-NNLS using an analyst-blinded, closed-library experimental data set obtained with a large-area plastic-scintillator RPM detector. Am-241, Cs-137, and Co-60 were selected from the sealed sources available at the test facility to represent low-, intermediate-, and high-energy gamma-ray responses relevant to standardized RPM performance evaluation. One fixed reference spectrum for each radionuclide and a measured background spectrum were used to analyze independently prepared single- and mixed-source measurements. Raw-spectrum NNLS and detector-specific five-window NNLS were applied as comparison methods under matched preprocessing and reference conditions. The evaluation focuses on constituent ranking, reconstruction residuals, and counting-statistical stability under static, unshielded conditions; formal compliance testing, direct activity quantification, open-set identification, and operational performance under shielding, source motion, cargo, or changing backgrounds are beyond the scope of this study.

Results

Single-radionuclide responses and distance dependence

The single-radionuclide measurements exhibited the broad continuum responses characteristic of plastic scintillation detectors. Channels 1–3 were below the lower-level discriminator and contained no usable spectral information; therefore, Channels 4–256 were retained for all subsequent transformations and fitting analyses. After channel-wise background subtraction, the total count rate integrated over Channels 4–256 decreased consistently with increasing source-to-detector distance for each radionuclide. In all nine non-reference single-source measurements, the coefficient corresponding to the correct radionuclide was the largest in the BCCS-NNLS fit.

Across all 12 single-radionuclide measurements, the normalized target-radionuclide coefficient closely tracked the normalized background-subtracted total count rate (Fig. 1b). Linear regression yielded $y=0.0181+0.9749x$, with $R^2=0.99986$, indicating that the fitted target coefficient consistently reflected the relative change in the measured source response. Because both quantities were derived from the same spectra and normalized to the same radionuclide-specific reference measurements, this relationship should be interpreted as an internal consistency check rather than as an independent calibration.

Parametric Poisson resampling yielded 100% correct-top-rank stability for nine of the 12 single-radionuclide measurements. Reduced stability was observed only for the weakest measurement of each radionuclide: 96.86% for Am-241 at 460 cm and 99.84% for both Cs-137 at 390 cm and Co-60 at 470 cm. As the net source response decreased and the relative influence of counting fluctuations and channel-wise background subtraction increased, small positive coefficients assigned to non-target radionuclides became more prominent, and the relative BCCS reconstruction residuals tended to increase. Thus, a correct top-ranked coefficient supports identification of the dominant radionuclide but does not by itself establish that all smaller non-target coefficients are statistically distinguishable from zero.

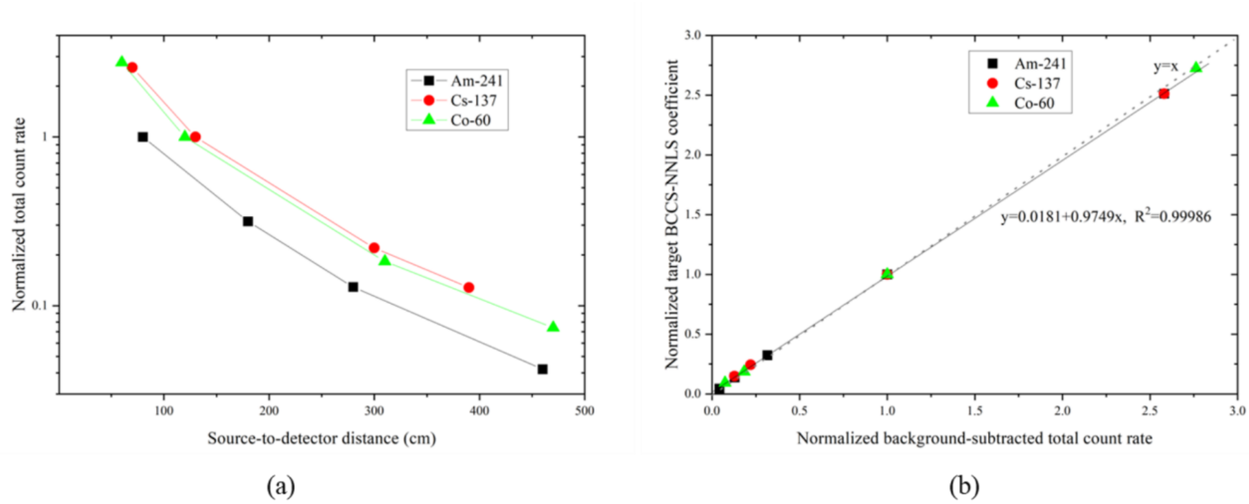

Figure 1. Distance-dependent response and consistency between total-count scaling and BCCS-NNLS coefficients for the single-radionuclide measurements. (a) Normalized background-subtracted total count rate as a function of source-to-detector distance for Am-241, Cs-137, and Co-60. The total count rate was integrated over Channels 4–256 and normalized to the corresponding radionuclide reference measurement: Am-241 at 80 cm, Cs-137 at 130 cm, and Co-60 at 120 cm. Lines connecting the measurements are provided only as guides to the eye. (b) Relationship between the normalized background-subtracted total count rate and the normalized target-radionuclide BCCS-NNLS coefficient for all 12 single-radionuclide measurements. The three reference measurements overlap at (1,1). The dashed gray line represents ideal proportionality (y=x), and the solid black line represents the linear regression, y=0.0181+0.9749x (R^2=0.99986).

The measured distance dependence deviated from the inverse-square relationship, particularly at short source-to-detector distances. The $1/r^2$ approximation treats the detector as point-like and is valid primarily under far-field conditions. In contrast, the detector used in this study had an active area of approximately 30×180 cm^2 and subtended a substantial solid angle at short distances. The measured responses were therefore compared with both the inverse-square approximation and a finite rectangular-detector solid-angle model [15].

For each radionuclide, the normalized measured response was defined as the background-subtracted total count rate integrated over Channels 4–256 divided by that of the corresponding reference measurement. The finite-detector estimates were calculated from the solid angle subtended by a centered rectangular detector with an active area of 30×180 cm^2 and were normalized to the same radionuclide-specific reference distance.

Table 1. Comparison of normalized measured responses with inverse-square and finite rectangular-detector solid-angle estimates.

Radionuclide	Distance (cm)	Inverse-square estimate	Finite-detector solid-angle estimate	Normalized measured response	Top-rank stability (%)
Am-241	80	1.000	1.000	1.000	100.00
	180	0.198	0.269	0.316	100.00
	280	0.082	0.118	0.129	100.00
	460	0.030	0.045	0.042	96.86
Cs-137	70	3.449	2.545	2.581	100.00
	130	1.000	1.000	1.000	100.00
	300	0.188	0.220	0.220	100.00
	390	0.111	0.132	0.128	99.84
Co-60	60	4.000	2.728	2.763	100.00
	120	1.000	1.000	1.000	100.00
	310	0.150	0.181	0.183	100.00
	470	0.065	0.081	0.074	99.84

Measured responses were normalized to the measurements obtained at 80 cm for Am-241, 130 cm for Cs-137, and 120 cm for Co-60. The inverse-square and finite-detector estimates were normalized to the same radionuclide-specific reference distances. The finite-detector estimates were calculated from the solid angle subtended by a centered rectangular detector with an active area of approximately 30×180 cm^2.

As shown in Table 1, the measured responses generally followed the finite-detector solid-angle estimates more closely than the inverse-square approximation. The Cs-137 responses differed from the finite-detector estimates by no more than approximately 10%, whereas deviations of up to approximately 20% were observed for Am-241 and Co-60. These differences are reasonable because the rectangular solid-angle model accounts only for the principal geometric effect and assumes a centered point source and a uniformly responding planar detector. It does not account for energy-dependent attenuation and scattering in the detector enclosure, detector-response non-uniformity, source-positioning uncertainty, background mismatch, or threshold effects. For Am-241, attenuation in the detector enclosure may be particularly relevant because its low-energy photon response is more sensitive to intervening structural materials. At short source-to-detector distances, photons reaching off-axis regions of the large detector face enter the enclosure at more oblique angles and therefore traverse a greater effective material thickness. Such near-field attenuation could suppress the response at the closest Am-241 reference position, causing the normalized responses at larger distances to appear higher than predicted by the idealized solid-angle model. At the largest distances, particularly for the weakest measurements, counting fluctuations and background subtraction may also contribute to the remaining deviations. These effects were not independently quantified in the present experiment.

Composition-blinded BCCS decomposition of mixed-source spectra

The primary BCCS analysis was performed using coded mixed-source spectra before their constituent identities and detailed measurement configurations were disclosed to the analyst. The coefficients of all three radionuclides in the reference library were estimated simultaneously by NNLS without supplying either the number or identities of the constituents in each test spectrum. The fitted coefficients and their rankings were finalized before the sample-composition key was disclosed. Agreement between the coefficient rankings and the actual constituent sets was then evaluated retrospectively.

For all eight binary mixtures, the two actual constituent radionuclides occupied the two largest BCCS-NNLS coefficient ranks. Sample 17 contained all three radionuclides in the reference library and returned positive coefficients for Am-241, Cs-137, and Co-60. Because the number of actual constituents in Sample 17 equaled the size of the reference library, this measurement was treated as a decomposition-consistency case rather than as a discriminative top-**K** test.

Figure 2 shows the measured and reconstructed BCCS profiles for all nine mixed-source measurements. The total NNLS reconstructions closely followed the measured background-subtracted BCCS profiles over the analyzed threshold-channel range, with relative residuals ranging from 0.48% to 1.31%. This agreement was maintained for both the same-distance configurations of Samples 13–17 and the source-dependent distance configurations of Samples 18–21.

For Sample 13, which contained Cs-137 and Co-60, the fitted BCCS coefficients were 0.032 for Am-241, 0.949 for Cs-137, and 0.797 for Co-60, with a relative residual of 0.48%. The corresponding conditional 95% parametric Poisson-resampling intervals were 0.000–0.066, 0.924–0.974, and 0.775–0.820, respectively. The coefficients of the two actual constituents were therefore clearly separated from the small off-target Am-241 coefficient.

For Sample 20, which contained Am-241 and Cs-137, the fitted coefficients were 0.946 for Am-241, 0.609 for Cs-137, and 0.019 for Co-60, with a relative residual of 0.87%. The corresponding conditional 95% parametric Poisson-resampling intervals were 0.913–0.979, 0.588–0.629, and 0.000–0.038, respectively. This result similarly preserved the correct top-two ranking while assigning only a small coefficient to the absent Co-60 component.

Small positive coefficients assigned to absent radionuclides were retained rather than manually set to zero because they provide information on spectral overlap, counting fluctuations, and mismatch between the measured response and the fixed reference library. Such off-target coefficients should not automatically be interpreted as evidence for additional radionuclides. Furthermore, the fitted coefficients are detector-response scaling factors relative to the fixed single-radionuclide references and cannot be converted directly into activity fractions without radionuclide-, geometry-, and detector-specific calibration and validated decision criteria.

Figure 2. Composition-blinded BCCS-NNLS decompositions of the nine mixed-source spectra. Panels (a–i) correspond to Samples 13–21, respectively. Each panel shows the measured background-subtracted BCCS profile, total NNLS reconstruction, and fitted Am-241, Cs-137, and Co-60 contributions over threshold Channels 4–256. Samples 13–17 were measured with the constituent sources positioned at the same source-to-detector distance, whereas Samples 18–21 used source-dependent distances, as indicated in the panel titles. The sample compositions shown in the panel titles were added after disclosure of the sample-composition key; they were not provided as inputs to the primary BCCS analysis. The fitted coefficients are detector-response scaling factors relative to the fixed single-radionuclide references and do not represent activity fractions. Small off-target components were retained without manual truncation. The reported relative residuals were calculated in the BCCS representation space.

Retrospective comparison with raw-spectrum and five-window NNLS

Having established the composition-blinded decomposition performance of BCCS-NNLS, raw-spectrum NNLS and detector-specific five-window NNLS were retrospectively applied to the same evaluation spectra using the same fixed radionuclide references and matched live-time normalization. For each method, the coefficients of all three radionuclides were estimated without supplying the actual sample composition. The disclosed composition key was used only to evaluate the resulting coefficient rankings using the known value of **K.**

The raw-spectrum method fitted each gross measurement using background-subtracted radionuclide templates together with an explicit background template. In contrast, BCCS-NNLS and five-window NNLS were applied to channel-level background-subtracted data represented in the BCCS and five-window spaces, respectively.

Across the 18 evaluation spectra—nine non-reference single-radionuclide measurements, eight binary mixtures, and one ternary mixture—all three methods produced coefficient rankings consistent with the disclosed constituent sets. For each non-reference single-radionuclide measurement, the actual radionuclide had the largest coefficient. For each binary mixture, the two actual radionuclides occupied the two largest coefficient ranks. Sample 17 was evaluated separately as a decomposition-consistency case because it contained all three radionuclides in the available reference library.

For the nine mixed-source spectra, the relative residuals ranged from 0.48% to 1.31% for BCCS-NNLS, from 0.88% to 1.60% for raw-spectrum NNLS, and from 0.46% to 2.57% for five-window NNLS. BCCS-NNLS produced a lower residual than five-window NNLS for seven of the nine mixtures, whereas five-window NNLS produced a lower residual for Samples 15 and 20. Raw-spectrum residuals were below 2% for all nine mixed-source measurements.

These residuals were calculated in different representation spaces and therefore should not be compared as directly interchangeable statistical measures. In particular, the raw-spectrum residual was calculated from a gross spectrum that could be dominated by the fitted background component at weak source levels, whereas the BCCS and five-window residuals were calculated from background-subtracted representations. Consequently, a low raw-spectrum residual may partly reflect accurate reconstruction of the background rather than only the net radionuclide response.

Table 2. Retrospective cross-method comparison for the nine mixed-source spectra. Relative residuals were calculated separately in the BCCS, original-channel, and five-window representation spaces and should not be interpreted as identical statistical quantities. The sample-composition key was used only to evaluate the fitted coefficient rankings.

Sample	Actual constituents	R_{net} (%)	BCCS residual (%)	Raw residual (%)	Window residual (%)	Ranking outcome
13	Cs-137 + Co-60	89.72	0.48	1.60	1.12	Correct top-2 for all methods
14	Am-241 + Co-60	13.53	1.31	0.88	2.57	Correct top-2 for all methods
15	Cs-137 + Co-60	16.85	1.20	0.99	1.02	Correct top-2 for all methods
16	Am-241 + Cs-137	44.56	0.53	0.97	1.43	Correct top-2 for all methods
17	Am-241 + Cs-137 + Co-60	24.68	1.31	1.16	2.41	All three coefficients positive for all methods
18	Co-60 + Am-241	38.23	0.78	1.11	1.46	Correct top-2 for all methods
19	Cs-137 + Co-60	20.45	1.12	0.96	1.14	Correct top-2 for all methods
20	Am-241 + Cs-137	54.82	0.87	1.15	0.46	Correct top-2 for all methods
21	Cs-137 + Co-60	29.65	1.30	1.11	1.62	Correct top-2 for all methods

The effect of net signal strength was additionally examined using R_{net}, the net-to-background count-rate ratio. Correct coefficient ranking was retained over the investigated signal-strength range, including the two weakest mixed-source measurements at R_{net} = 13.53% and 16.85%. Nevertheless, weaker source responses generally showed greater sensitivity to counting fluctuations and background subtraction. No signal-strength cutoff was optimized from the present limited data set.

Performance may also deteriorate under conditions not represented in the present measurements, including gain shifts, background changes, altered source geometry, shielding, or radionuclides absent from the fixed reference library.

Complete raw-spectrum and five-window NNLS coefficients and rankings for all source measurements are provided in Supplementary Table S2b.

Methods

Experimental system and measurement geometry

Measurements were performed using a cRPM-54 portal-monitor testbed equipped with a large-area plastic scintillation panel measuring 30 × 180 × 5 cm^3 and optically coupled to a photomultiplier tube. The high-voltage supply and pulse-height acquisition system was a custom M1 International unit incorporating an Amptek DP5G digital pulse processor. The photomultiplier high voltage was set to 1,475 V, the amplifier gain to 4.0, and pulse-height spectra were recorded in 256 channels.

Sealed point sources of Am-241, Cs-137, and Co-60 were used, with nominal activities of 90, 16, and 9 µCi, respectively. The front surface of the detector panel was defined as the origin of the source-to-detector distance, $r = 0$. Each source was positioned on the normal axis passing through the center of the active detector area, with the source height aligned with the detector-center height.

For Samples 13–17, all constituent sources were placed at the same nominal source-to-detector distance. For Samples 18–21, the constituent sources were placed at different distances, while each source remained aligned with the detector center. No intentional shielding, cargo, or other intervening material was placed between the sources and the detector.

The primary data set therefore represent static, unshielded conditions using sealed point sources and a single large-area detector panel. They do not reproduce vehicle occupancy, cargo attenuation, shielding, complex load-dependent scattering, moving-source profiles, or time-varying environmental backgrounds encountered in operational portal monitoring. The experiment should accordingly be interpreted as a controlled feasibility evaluation specific to the detector configuration, acquisition settings, and radionuclide reference library used in this study.

The experimental arrangement and the single-radionuclide responses used to construct the NNLS reference library are shown in Fig. 3. Before spectral processing, each reference spectrum was converted to count rate using its file-specific MCA live time.

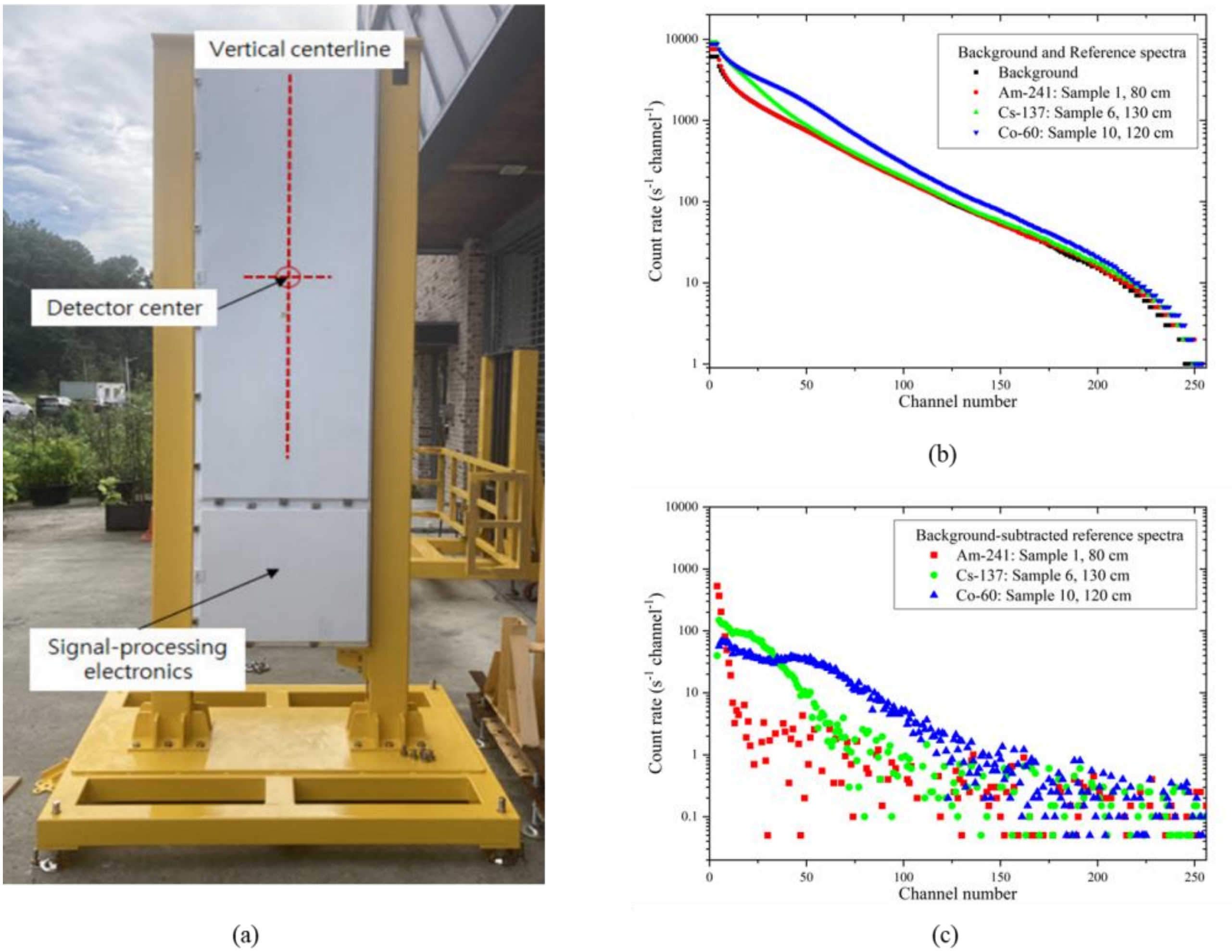

Figure 3. Experimental geometry and fixed single-radionuclide reference spectra.
(a) Front photograph of the cRPM-54 detector panel. The dashed line indicates the vertical centerline of the active detector area, and the marked intersection identifies the detector center and projected source position. The lower enclosure houses the high-voltage, signal-processing, and data-acquisition electronics.
(b) Live-time-normalized gross spectra of the measured background and the fixed Am-241, Cs-137, and Co-60 reference measurements.
(c) Corresponding background-subtracted reference spectra used to construct the radionuclide response library. The reference measurements were Am-241 at 80 cm, Cs-137 at 130 cm, and Co-60 at 120 cm.

Data set, acquisition time, and channel treatment

The complete experimental data set comprised 22 spectra: one independently acquired background spectrum and 21 numbered source measurements. The source measurements consisted of twelve single-source spectra (Samples 1–12) and nine mixed-source spectra (Samples 13–21). The radionuclide compositions, source-to-detector distances, and source arrangements are summarized in Supplementary Table S1.

All acquisitions had a nominal real-time duration of 20.0 s. The actual MCA live time recorded in the file headers ranged from 18.672 to 20.0 s. Each channel count was therefore divided by its file-specific live time before spectral comparison or decomposition. The background acquisition had a live time of 20.0 s and contained 121,966 counts over Channels 4–256, corresponding to a total background count rate of 6,098.3/s. For the primary detector panel, Channels 1–3 were below the lower-level discriminator and were excluded, leaving Channels 4–256 for analysis. In the supplementary validation using the opposing panel, only Channels 1 and 2 contained zero counts; therefore, Channels 3–256 were retained for the panel-specific BCCS analysis. The BCCS calculation was applied over this complete valid range rather than over a separately selected low-channel interval. Although the largest differences among the reference BCCS profiles occurred at relatively low threshold channels, each BCCS coordinate included all counts from its threshold channel to Channel 256.

No traceable channel-to-energy calibration was established for the present data set. The plastic-scintillator spectra did not contain narrow full-energy peaks suitable for conventional centroid and full-width-at-half-maximum calibration; consequently, the analysis was performed in detector-channel space. This avoids assigning unsupported physical energies to the channels but makes the measured reference library and the five-window boundaries specific to the present detector gain, lower-level discriminator setting, and electronics configuration. New reference measurements or recalibration would therefore be required after changes in gain, discriminator setting, or pulse-processing electronics.

One fixed single-radionuclide response was selected for each library member: Sample 1 for Am-241 at 80 cm, Sample 6 for Cs-137 at 130 cm, and Sample 10 for Co-60 at 120 cm. The remaining eighteen source spectra were used as evaluation measurements. Their compositions and detailed measurement conditions were concealed during the primary BCCS analysis and disclosed after the predictions had been finalized. The fitted coefficients are response-scaling factors relative to the selected reference measurements. They are not activity fractions because the source activities, distances, geometries, emission characteristics, attenuation conditions, and energy-dependent detector efficiencies differed among measurements.

Finite-detector solid-angle estimate

For comparison with the measured distance dependence, the detector was approximated as a centered rectangular plane with half-width a = 15 cm and half-height b = 90 cm. For an isotropic point source located on the central normal axis at a perpendicular distance r from the detector front surface, the solid angle subtended by the rectangular detector was calculated as

$$\Omega(\mathrm{r}) = 4tan^{-1}\left[\frac{ab}{r\sqrt{r^2+a^2+b^2}}\right]$$

The normalized finite-detector estimate for each radionuclide was calculated as

$$\mathrm{G(r)} = \frac{\Omega(r)}{\Omega(r_{ref})}$$

where r_{ref} was the distance of the selected reference measurement for that radionuclide. This calculation assumes an isotropic point source, a centered planar detector, and a detector response proportional to the subtended solid angle. It does not include attenuation, scattering, detector thickness, edge effects, or energy-dependent detection efficiency.

Backward cumulative channel sum and background compensation

Let C_i denote the live-time-normalized count rate in channel i and let N = 256. The BCCS at threshold i is defined as

$$B_i = \sum_{j=i}^{N} C_j \qquad (1)$$

Denoting the BCCS transformation by **T**, its linearity satisfies

$$\mathrm{T}(C_1 + C_2) = \mathrm{T}(C_1) + \mathrm{T}(C_2).$$

Background correction was therefore performed without altering the additive mixture structure:

$$S^M = \mathrm{T}(C^M) - T(C^{bg}) = T(C^M - C^{bg}) \qquad (2)$$

where C^M and C^{bg} are the live-time-normalized measurement and background spectra, respectively. Negative channel-level net values produced by counting fluctuations were retained before application of the BCCS transformation rather than being clipped to zero. The resulting BCCS measurement vector contained all threshold coordinates from Channel 4 through Channel 256.

A background-ratio representation,

$$\frac{T(C^M)}{T(C^{bg})} - 1,$$

was also examined as a visualization because it can increase apparent contrast when the background is stable. However, uncertainty in the denominator can amplify fluctuations at weak high-channel count levels. The ratio representation was therefore not used to calculate coefficients, reconstruction residuals, ranking statistics, or signal-strength criteria.

NNLS decomposition, ranking rule, and residual

The background-subtracted BCCS vector **y** was modeled using a reference matrix **A**, whose columns were the Am-241, Cs-137, and Co-60 BCCS reference profiles. The coefficient vector was estimated by NNLS [16]:

$$\hat{a} = \underset{a \geq 0}{\arg min} \parallel \boldsymbol{y} - \boldsymbol{A}\boldsymbol{a} \parallel_2^2 \tag{3}$$

The representation-specific relative reconstruction residual was defined as

$$\varepsilon = 100 \times \frac{\|\boldsymbol{y}-\boldsymbol{A}\hat{\boldsymbol{a}}\|_2}{\|\boldsymbol{y}\|_2} \tag{4}$$

The NNLS calculation estimated all three radionuclide coefficients and did not require the number of constituents as an input. During the primary BCCS analysis, the sample compositions and detailed measurement conditions were not disclosed. After the coefficient rankings had been finalized and the answer key released, constituent agreement was evaluated retrospectively.

For a single-source measurement, agreement was recorded when the actual radionuclide had the largest coefficient. For a binary mixture, agreement was recorded when the two actual constituents occupied the two largest coefficient ranks. For ternary Sample 17, the number of actual constituents equaled the size of the three-radionuclide library. Positive recovery of all three coefficients was therefore treated as a decomposition-consistency result rather than as a discriminative top-**K** test. No open-set, unknown-**K**, or radionuclide-specific decision thresholds were validated.

Adjacent BCCS threshold coordinates contain many of the same channel counts and therefore have correlated uncertainties. The unweighted BCCS residual quantifies reconstruction agreement within cumulative-response space, but it is not a chi-square probability and does not represent independent statistical evidence at every threshold coordinate.

Raw-spectrum NNLS baseline

As a full-channel comparison baseline, the original live-time-normalized gross spectrum was fitted over Channels 4–256 without prior background subtraction. The reference matrix contained the measured background template b and the three background-subtracted single-radionuclide responses $\mathbf{s}_x$:

$$y^{raw} \approx \beta\boldsymbol{b} + a_{Am}\boldsymbol{s}_{Am} + a_{Cs}\boldsymbol{s}_{Cs} + a_{Co}\boldsymbol{s}_{Co} + \boldsymbol{e} \tag{5}$$

where β is the fitted background-scaling coefficient and ***e*** is the reconstruction error. All coefficients were constrained to be non-negative. The background coefficient β was excluded from the radionuclide ranking, and the relative residual was calculated in the original 253-channel space.

Because the gross spectrum is dominated by environmental background at weak source levels, a small raw-spectrum residual may primarily indicate an accurate fit of the background component. It does not by itself demonstrate reliable radionuclide discrimination.

Detector-specific five-window NNLS baseline

A low-dimensional five-window baseline was implemented according to the established principle that energy-window boundaries for PVT detectors should be selected for the specific detector response, gain configuration, and target radionuclide library rather than transferred directly from another RPM system [3,4]. The window boundaries were selected from the three single-radionuclide reference spectra and fixed before application to the evaluation spectra.

The same file-specific live-time normalization and channel-wise background subtraction used for BCCS were applied before window integration. The net count rates were then summed within the five fixed intervals to form a five-element measurement vector. The corresponding five-element Am-241, Cs-137, and Co-60 reference vectors were fitted by NNLS.

Because an independently measured background spectrum had already been subtracted before window integration, W5 was not used to estimate a separate background coefficient. It instead served as a high-channel control interval sensitive to residual background mismatch and high-energy spectral contributions. The residual for this method was calculated in the five-dimensional window-response space.

Table 3. Detector-specific channel windows used for the five-window NNLS baseline.

Window	Channel range	Number of channels	Primary role
W1	4-10	7	Am-241-dominant low-channel region
W2	11-25	15	lower Cs-137-sensitive region
W3	26-40	15	upper Cs-137-sensitive and transition region
W4	41-80	40	Co-60-sensitive region
W5	81-256	176	High-channel control region sensitive to residual background mismatch

The corresponding five-window reference-response vectors are provided in Supplementary Table S4.

Signal-strength measure and empirical operating region

The net source contribution relative to the independently measured background was expressed as

$$R_{net} = 100 \times \frac{C_{sample} - C_{background}}{C_{background}} \qquad (6)$$

where C_{sample} and $C_{background}$ are the gross sample and background count rates, respectively, integrated over Channels 4–256.

Thus, R_{net} = 20% corresponds to a gross sample count rate equal to 120% of the measured background count rate. For conservative practical interpretation of the present measurements, $R_{net} \geq 20\%$ was considered a relatively strong-signal region. This value was not derived as an optimized classification threshold and should not be interpreted as a detection limit or alarm criterion.

Parametric counting-statistical resampling

Counting-statistical stability was evaluated using 5,000 parametric Poisson resamples for each evaluation spectrum. For each resample, the observed count in every sample channel and every background channel was used as the mean of an independent Poisson distribution. The resampled channel counts were divided by their original file-specific live times, background-subtracted channel by channel, transformed by BCCS, and fitted against the fixed measured reference matrix.

The resulting coefficient distributions were summarized using the median and the 2.5th and 97.5th percentiles. For single-source measurements, top-rank stability was defined as the fraction of resamples in which the actual radionuclide produced the largest coefficient. For binary mixtures, constituent-ranking stability was defined as the fraction of resamples in which the two actual radionuclides occupied the two largest coefficient positions. For ternary Sample 17, the fraction of resamples in which all three coefficients were positive was reported as a decomposition-consistency measure rather than as a formal identification criterion.

The number of constituents used in these resampling metrics was obtained from the disclosed answer key and was used only for retrospective evaluation; it was not supplied to the NNLS fitting procedure.

ISO 11929-3:2025 requires a defined measurand, a complete uncertainty model, covariance treatment, and specified error probabilities for unfolding-based decision thresholds and detection limits [17]. The present resampling analysis was not intended to constitute such an assessment.

The experimentally measured reference matrix was treated as fixed in all resamples. The analysis therefore quantifies counting-statistical variation around the acquired sample and background spectra but does not include uncertainty arising from independent reference acquisition, source repositioning, certified source activity, detector gain drift, temperature variation, detector aging, electronics changes, or temporal and spatial variation in environmental background.

The complete coefficient distributions and ranking-stability results are summarized in Supplementary Table S3.

Discussion and conclusions

This study demonstrated that single- and mixed-radionuclide measurements can be decomposed using a measured background spectrum and fixed single-radionuclide reference spectra. Under stable detector and electronics conditions, changes in source intensity or distance are reflected primarily in the fitted coefficients, while the broad radionuclide-dependent response shapes remain sufficiently consistent for decomposition. The BCCS transformation preserves this linear relationship and enables reconstruction of a measured BCCS profile as a non-negative combination of the reference responses. A sufficient net source signal is still required, but the measurement intensity does not need to match that of the reference spectrum.

The three evaluated methods represent the same spectral data in different forms. Raw-spectrum NNLS retains all channels and explicitly models the background, but its residual can be strongly influenced by the background when the source contribution is weak. Five-window NNLS is simple and computationally efficient, but it depends on detector-specific window boundaries and discards spectral variation within each window. BCCS avoids reliance on a small set of fixed windows and instead produces a channel-resolved cumulative-response profile suitable for NNLS decomposition. Because the residuals were calculated in different representation spaces, their absolute values should be interpreted as method-specific reconstruction measures rather than directly comparable statistical quantities.

Following completion of the composition-blinded analysis and disclosure of the answer key, BCCS correctly ranked the actual constituents in all evaluated non-reference single-source and binary-mixture measurements in the primary-panel data set. The ternary measurement also returned positive contributions from all three reference radionuclides, and Poisson resampling indicated that the rankings were generally stable against counting fluctuations. The fitted coefficients should not be interpreted as activities or activity fractions because they also depend on distance, geometry, attenuation, emission characteristics, and detector efficiency. Likewise, the empirical R_{net} = 20% region is a quality-control indicator for the present detector and data set, rather than a universal decision threshold or formal detection limit.

Additional validation using the opposing detector panel, operated at a different high voltage and analyzed using its panel-specific valid channel range, background spectrum, and reference spectra, reproduced the correct coefficient rankings for all nine non-reference single-source measurements and all eight binary mixtures. All three coefficients were positive for the ternary mixture, and Poisson resampling likewise showed high ranking stability (Supplementary Note 1 and Tables S5–S7). These results support reproducibility across detector panels when detector-specific measurement characteristics and reference spectra are taken into account, but do not demonstrate calibration-free transfer of reference spectra between panels.

A major practical advantage of BCCS–NNLS is that it requires only channelized spectral data. An existing RPM or radiation monitor that already records pulse-height spectra could therefore apply the method without replacing the detector or adding separate spectroscopic hardware. The required processing is limited to background subtraction, cumulative summation, and estimation of a small number of non-negative coefficients. No model training, high-resolution peak reconstruction, or specialized computing hardware is required. In field use, the analysis could be initiated after a count-rate alarm to indicate which reference radionuclides are most consistent with the measured response. This first-stage information would not replace confirmatory identification, but it could help prioritize secondary inspections and support faster radiation-safety decisions.

The three radionuclides evaluated here span low-, intermediate-, and high-energy gamma-ray response regions relevant to RPM performance evaluation. However, the present static measurements with unshielded sealed point sources and a fixed detector configuration do not constitute formal compliance testing. The reference library was also limited to three radionuclides. Broader application will require additional detector-specific references and validation under source motion, shielding, cargo attenuation, background variation, gain drift, and source-position changes. Out-of-library radionuclides must also be considered because a closed-library model may assign an unknown response to the most similar available reference.

Overall, BCCS–NNLS provides a simple and interpretable method for closed-library radionuclide decomposition using a large-area plastic-scintillator detector. Its principal practical value is that it can be introduced primarily as a software upgrade when channelized spectral data are already available. With an expanded reference library and validation under realistic operating conditions, the method could support first-stage radionuclide screening in existing RPM systems.

Data availability

The data underlying this study are available in Zenodo at [DOI]. The repository includes the raw MCA files acquired with the primary and opposing detector panels. For each panel, the archived data comprise one background measurement, twelve single-source measurements, and nine mixed-source measurements, together with the corresponding metadata and processed analysis results.

Code availability

The Python analysis code used to reproduce the results reported in this study is available in Zenodo at [DOI]. The archive includes scripts for MCA data import, live-time normalization, background subtraction, BCCS transformation, raw-spectrum and five-window NNLS analyses, reconstruction-residual and signal-strength calculations, finite-detector solid-angle estimation, parametric Poisson resampling, and generation of the reported tables and figures. It also includes dependency information, parameter settings, fixed random seeds, and instructions for reproducing the analysis.

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Author contributions

M.M. conceived the BCCS-based radionuclide-identification approach, planned and supervised the study, designed and performed the experimental measurements, interpreted the results, and co-wrote the manuscript. J.-Y.S. developed the RPM data-acquisition software, implemented the BCCS- and NNLS-based analysis software, performed the spectral processing and quantitative analyses, interpreted the results, and co-wrote the manuscript. B.J. generated Monte Carlo-based plastic-scintillator spectra used during method development and preliminary validation, organized the evaluation data, and contributed to interpretation of the results. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

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